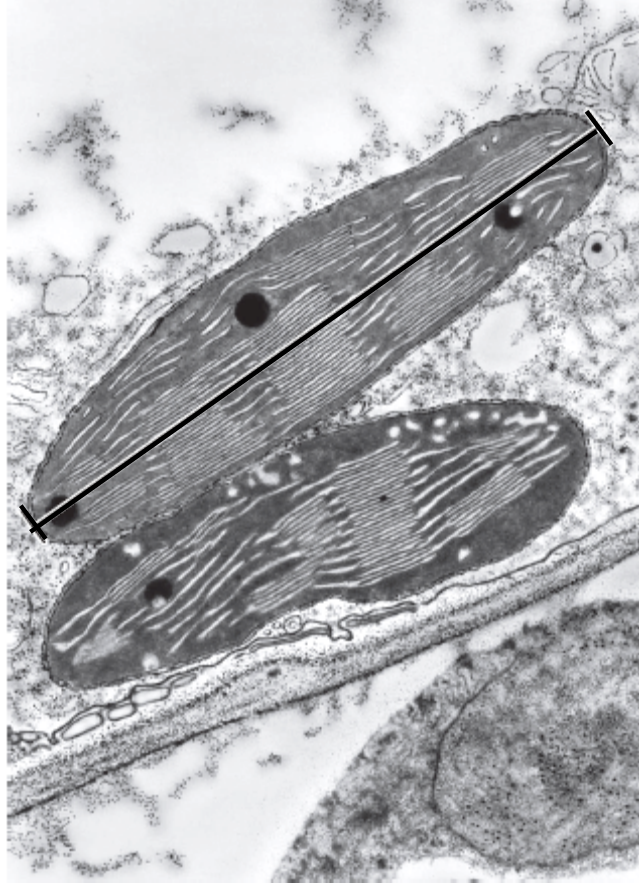


- 1 The electron micrograph shows a chloroplast.

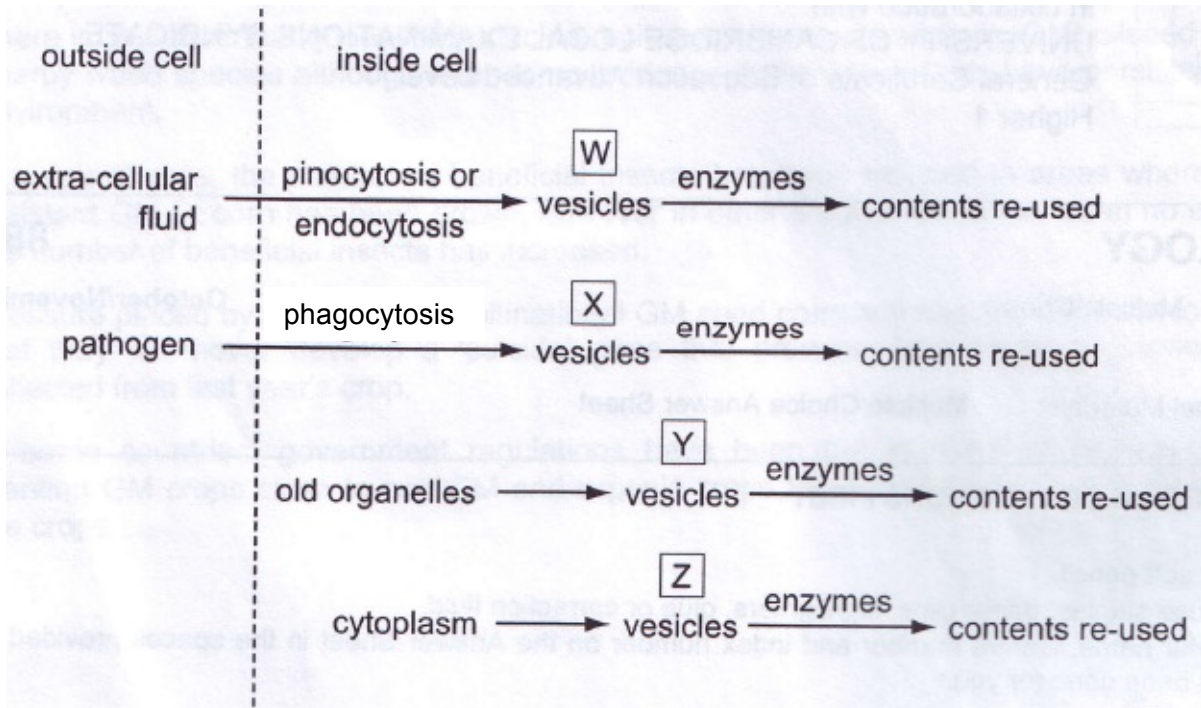


The length of the chloroplast along the line shown is 80 μm . The actual length of the chloroplast is 10 μm .

What is the magnification of the chloroplast?

- A** $\times 8 \times 10^2$ **B** $\times 8 \times 10^3$ **C** $\times 8 \times 10^4$ **D** $\times 8 \times 10^6$

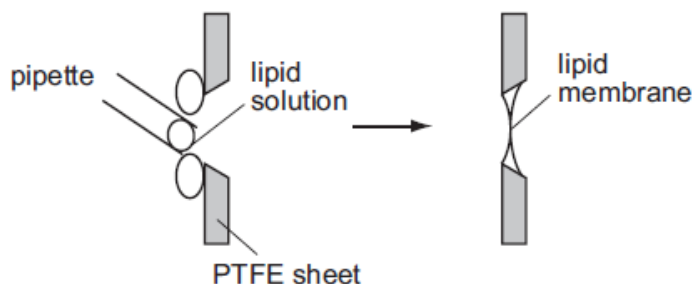
- 2 The flow chart shows processes which take place inside animal cells.



Which processes require the activity of lysosomes?

- A** W, X, Y and Z
- B** W and X only
- C** X and Y only
- D** Y and Z only

- 3 Lipid membranes can be formed in the laboratory by painting phospholipids over a PTFE sheet with a hole in it.



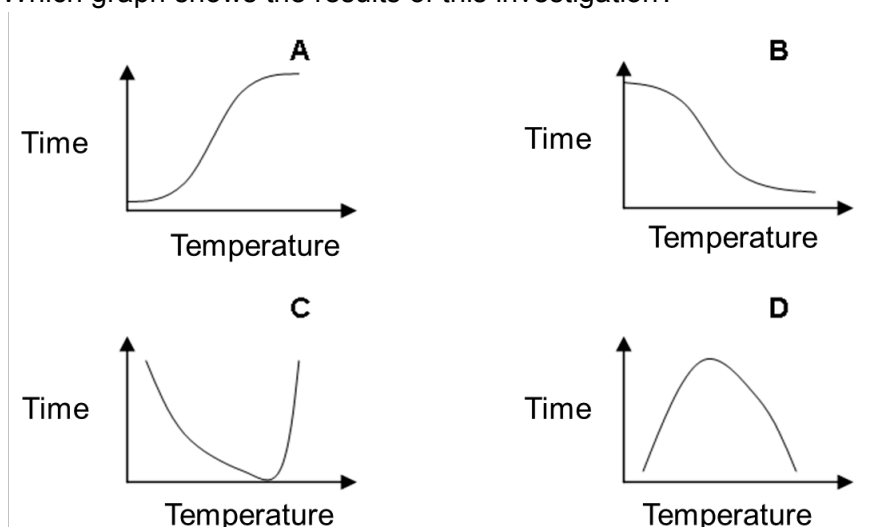
Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ . In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added, at which time current flowed.

Which statement is consistent with this information and your knowledge of membrane structure?

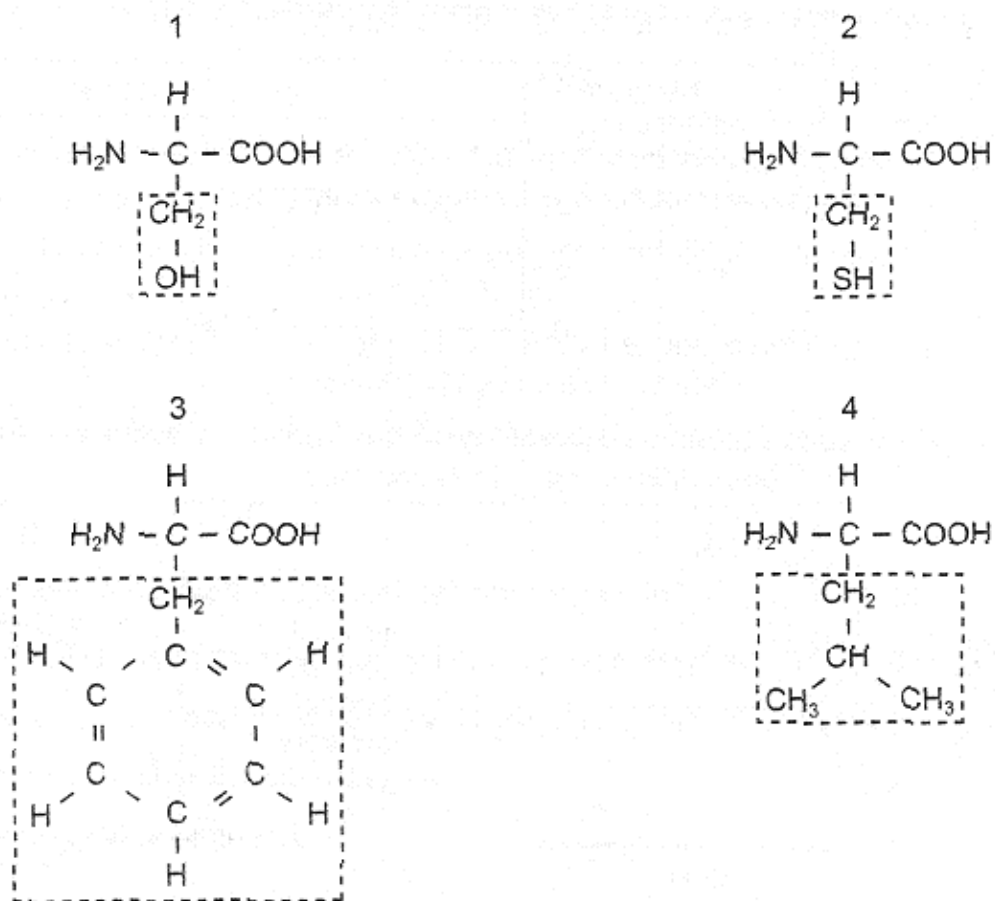
Gramicidin becomes incorporated into the membrane and is

- A a carbohydrate molecule found only on the outside of the membrane.
 - B a non-polar lipid which passes all the way through the membrane.
 - C a protein molecule with both hydrophilic and hydrophobic regions.
 - D a protein molecule which has only hydrophobic regions.
- 4 Iodine test gives a blue-black colour in the presence of starch. The time taken for the blue-black colour of a fixed amount of starch to disappear was recorded over a range of temperature.

Which graph shows the results of this investigation?



- 5 The diagram shows four different amino acids, each with a different R group (side chain).



Which amino acid could form a hydrophobic interaction between their R groups?

- A** 1 and 2 only
- B** 1 and 3 only
- C** 2 and 4 only
- D** 3 and 4 only

- 6 A living cell from the lung epithelium of a newt is shown at different stages of mitosis in the micrographs (A to F) below. Identify the correct sequence of these micrographs.

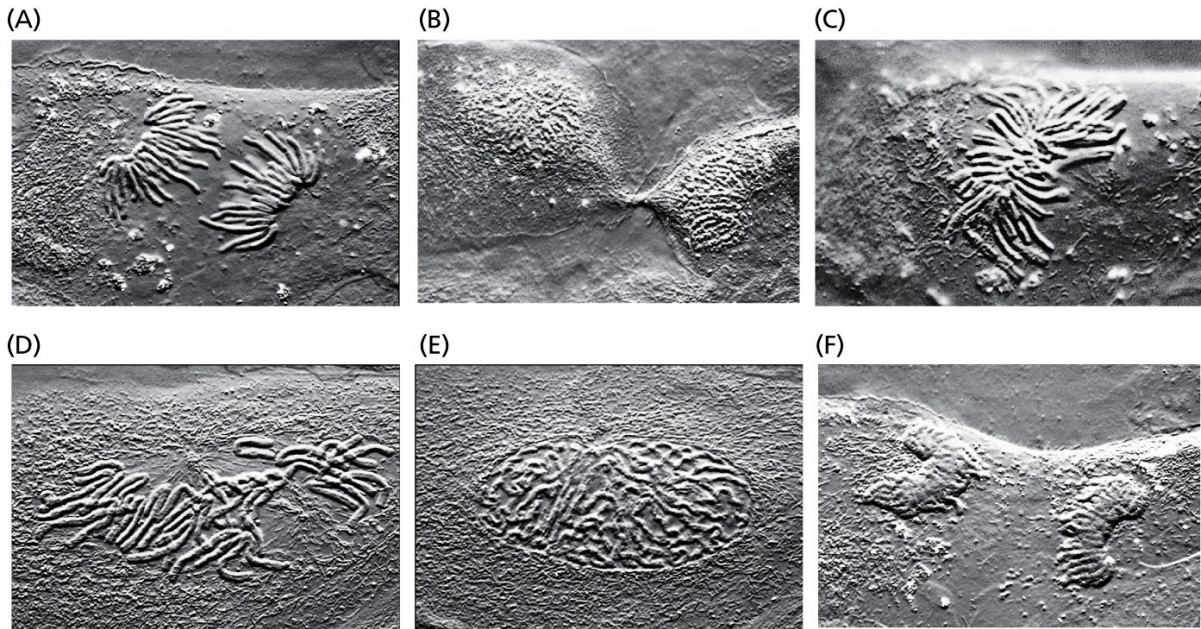


Figure 17-14 MBoC5: The Problems Book (© Garland Science 2008)

- A **DECAFB**
- B **EDCAFB**
- C **ECDAFB**
- D **EDCBAF**
- 7 Human immunodeficiency virus (HIV) is a retrovirus. After infecting a host cell, viral DNA is produced which is incorporated into the DNA of the host cell. The modified host genome now codes for the production of new HIV particles.

Which could be used as a potential treatment to slow down the spread of HIV?

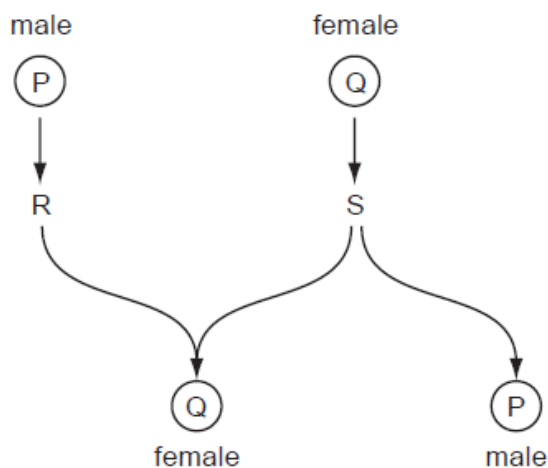
1. inhibitors of restriction endonucleases
2. inhibitors of reverse transcriptase
3. restriction endonucleases
4. reverse transcriptase

- A 1 only B 2 only C 1 and 4 only D 2 and 3 only

- 8 Which of the following shows the possible effects of a single nucleotide substitution in each of the following locations in a gene on the production of the protein it codes for?

	Promoter	Transcription terminator	Start codon	Stop codon	Middle of an intron
A	Protein product is longer than normal	Too much protein product is produced	Protein product is normal	Protein product is shorter than normal	No protein product is produced
B	No protein product is produced	Protein product is shorter than normal	Protein product is longer than normal	Protein product is normal	Too much protein product is produced
C	Protein product is normal	Protein product is longer than normal	Protein product is shorter than normal	Too much protein product is produced	Protein product is longer than normal
D	Too much protein product is produced	Protein product is normal	No protein product is produced	Protein product is longer than normal	Protein product is normal

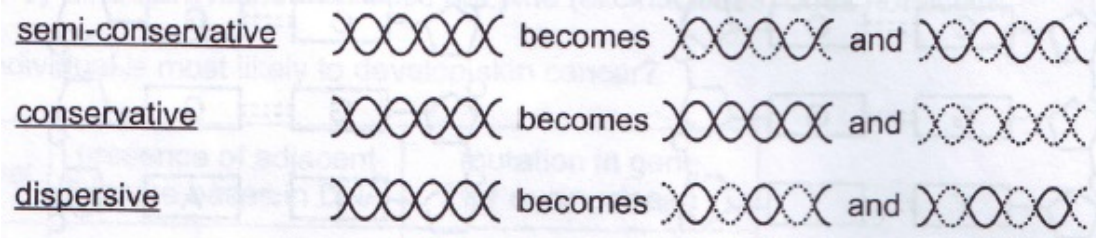
- 9 Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes.



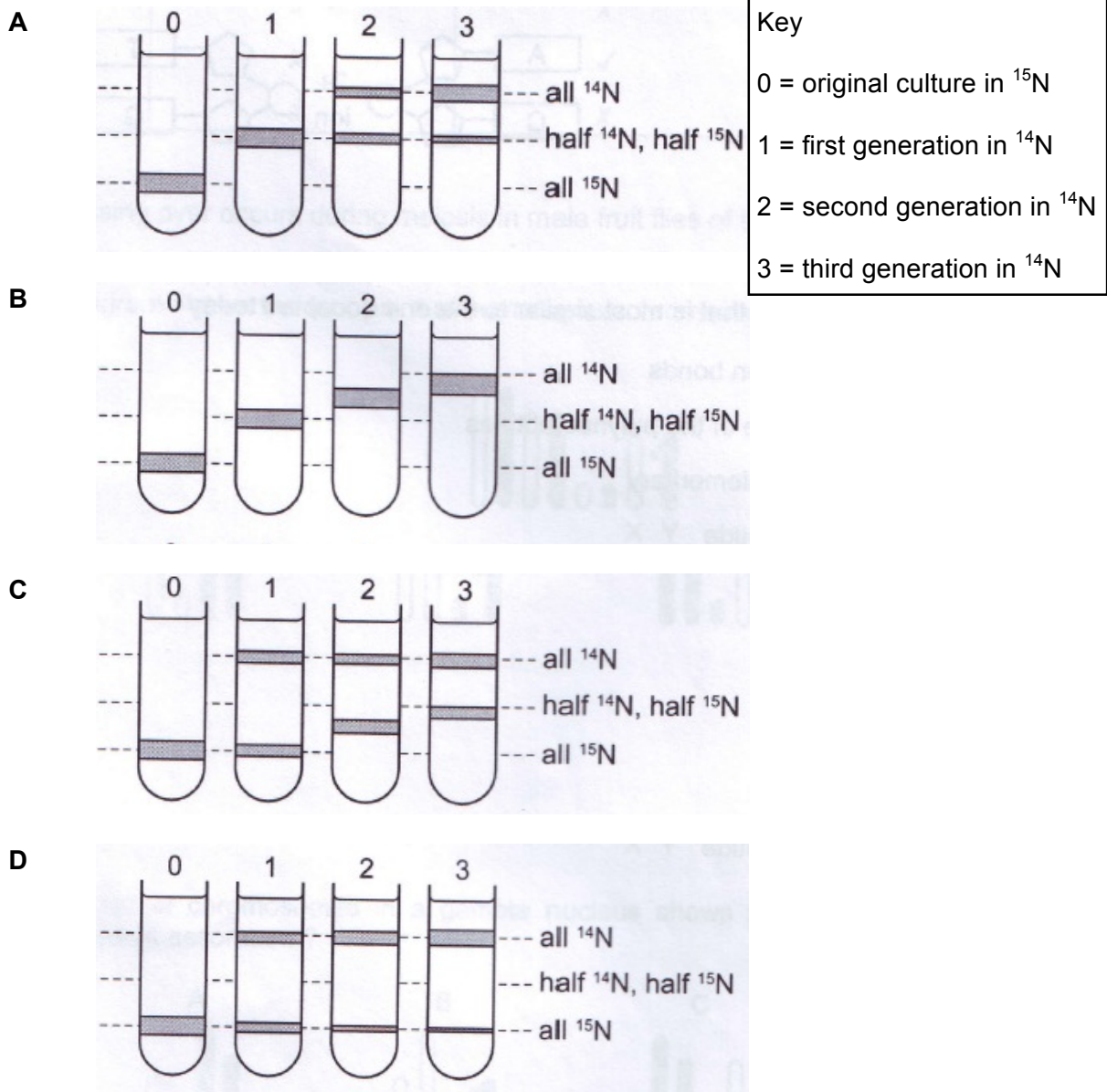
Using the diagram, which row in the table shows how sex is determined in these insects?

	P	Q	R	S
A	n	2n	mitosis	meiosis
B	n	n	mitosis	mitosis
C	2n	n	meiosis	meiosis
D	2n	2n	meiosis	mitosis

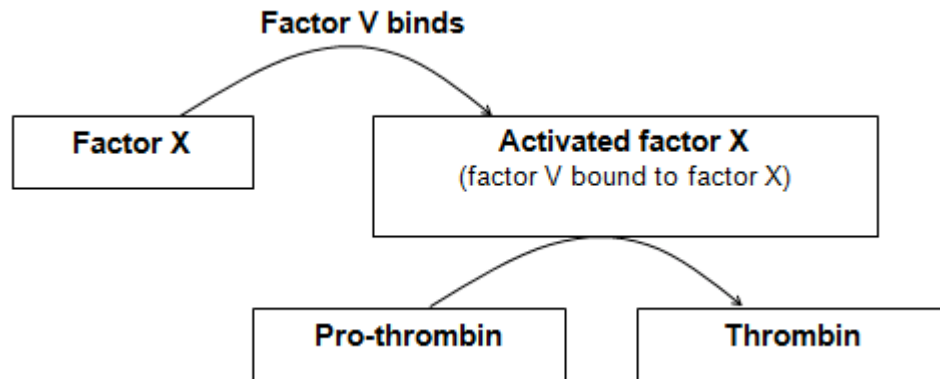
- 10 Meselson and Stahl found that in dividing cells, DNA is copied by semi-conservative replication. At the time of their discovery it was thought that DNA might be copied in one of three ways. In the diagram, the original DNA strands are shown by solid lines and the copy strands by dotted lines.



Which set of results would have proved that the DNA replication was **conservative**?



- 11 Multiple proteins are involved in the process of blood clotting. The diagram below shows how factor V functions as a co-factor in blood clotting.



Thrombin goes on to cleave fibrinogen to form fibrin, which helps in blood clot formation. Another protein that is involved in blood clotting is activated protein C (aPC), a natural anticoagulant that acts to limit the extent of clotting by cleaving and degrading factor V. A mutation in factor V protein results in a variant protein called factor V Leiden. The Leiden variant of the protein is resistant to degradation by aPC.

Which of the following option shows the mode of inheritance of the Leiden allele, the correct type of mutation this is and the result this mutation will have on blood clotting?

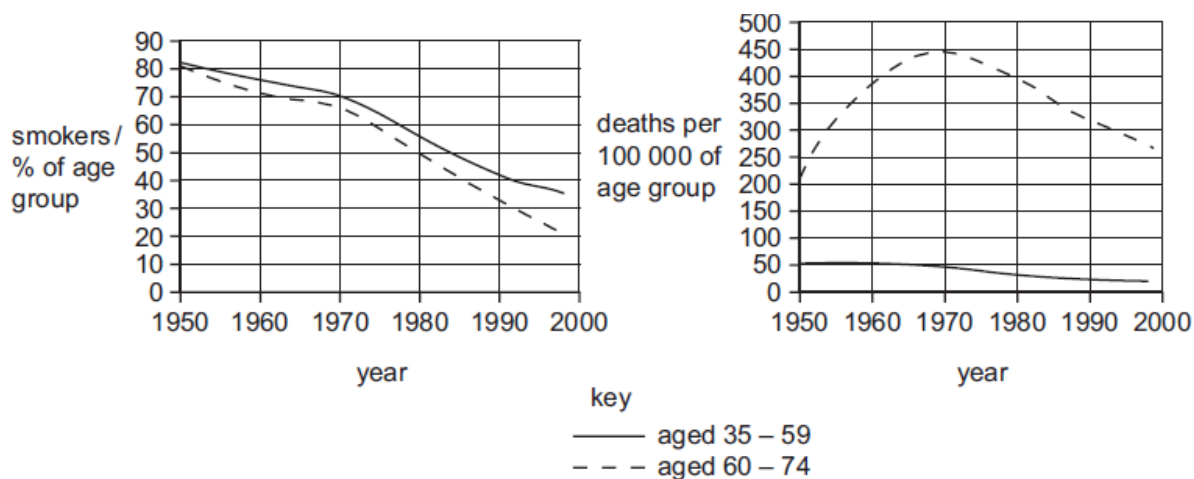
	Mode of inheritance	Type of mutation	Result on blood clotting
A	Dominant	Gain-of-function	Blood clots more easily
B	Dominant	Loss-of-function	Blood clots less easily
C	Recessive	Gain-of-function	Blood clots less easily
D	Recessive	Loss-of-function	Blood clots more easily

- 12 Mature mRNA and DNA are extracted from the same cells and mixed in conditions that favour the mRNA and complementary parts of the DNA becoming linked. Some parts of the DNA remain as double-stranded DNA. Other parts of the DNA form equal numbers of single-stranded DNA loops and double-stranded DNA-mRNA loops.

What describes these loops?

- A** exons form single-stranded loops, introns form double-stranded loops
- B** exons form double-stranded loops, introns form single-stranded loops
- C** template exons form double-stranded loops, non-template exons form single-stranded loops
- D** template introns form double-stranded loops, non-template introns form single-stranded loops

- 13** Some studies suggest that smoking increases the risk of developing lung cancer. The two graphs show the percentage of smokers and the deaths from lung cancer in men of two age groups between 1950 and 1998.



Which statement is not supported by the data in the graphs?

- A** Deaths from lung cancer in men aged 35-59 decreased by 50% over the period of the study.
- B** Deaths from lung cancer in men aged 60-74 increased up to 1970.
- C** The data for men aged 60-74 between 1950 to 1970 suggests that lung cancer takes up to 20 years to develop.
- D** The number of men aged 35-59 who were smokers decreased by approximately 60% over the period of the study.
- 14** Which statement correctly describes a role of histone proteins?
- A** All eukaryotic genes are transcribed continuously because they are not packaged by histones.
- B** DNA must be selectively released from its histone packaging before transcription can occur in bacteria.
- C** Histones package prokaryote chromatin into the nucleosomes that form the bulk of the chromosome.
- D** The organisation of DNA by histones in eukaryotes allows some gene control sequences to be thousands of base pairs away from the gene concerned.

15 Which of the following statements about RNA splicing of a single pre-mRNA are correct?

1. Alternative splicing controls the amount of gene products formed by having different promoters.
2. Alternative splicing controls the type of gene products formed by having different exons.
3. Different gene products can be formed at the same time within the same cells.
4. Different gene products can be formed at different stages of an organism's life cycle.
5. Different mRNA transcripts can be produced at the same time.

- A** 2 and 4 only
B 3 and 5 only
C 1, 2 and 4 only
D 1, 3 and 5 only

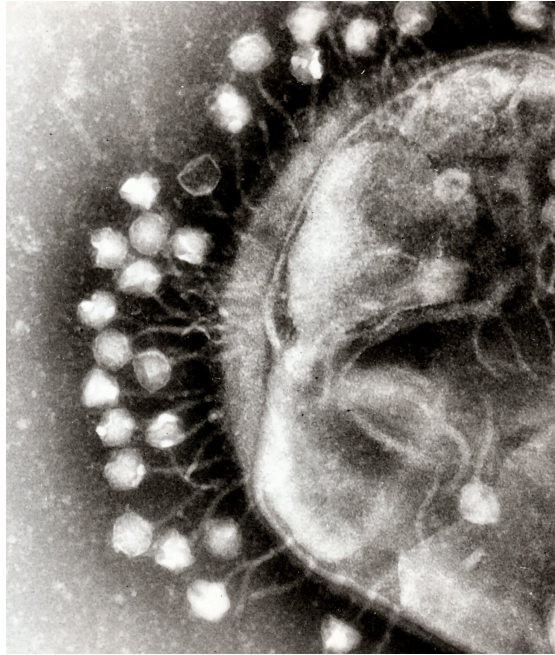
16 *Staphylococcus aureus* is a common bacterium, found on human skin. There are many strains of *S. aureus*. The antibiotic methicillin was used to treat infection by *S. aureus*. Now there are at least 15 different strains of MRSA (methicillin resistant *Staphylococcus aureus*).

Which of the following are valid reasons for the emergence of 15 different strains of MRSA?

1. The bacteria mutated when it was exposed to methicillin, thus becoming resistant.
2. The bacteria underwent spontaneous mutation and some strains happened to be resistant to methicillin.
3. The bacteria may have been exposed to a virus that carried a methicillin-resistant gene and underwent specialised transduction.
4. The antibiotic caused the bacteria to produce methicillin-resistant proteins.

- A** 1 and 2 only
B 2 and 3 only
C 1, 2 and 4 only
D 2, 3 and 4 only

17



With reference to the lambda phage shown in the diagram above, which of the following statements are **false**?

1. The genetic material of the virus can be either DNA or RNA.
2. The adsorption of viral particles to host cell is a chemical interaction in which strong covalent bonds are formed between the tail fibres of the virus and complementary receptor sites on the host cell wall.
3. Once the viral genetic material has reached the cytoplasm of the host cell, viral proteins will be degraded.
4. The host genes can be transferred by the virus to another host cell through the process of generalized transduction.
5. Mature infective virions have spontaneously assembled and are ready to leave host cell to infect other cells.
6. The virus is able to multiply via both the lytic and lysogenic cycles.

- A** 1 and 5 only
- B** 2 and 4 only
- C** 1, 2 and 5 only
- D** 1, 5 and 6 only

- 18** Ribonuclease is an enzyme that digests RNA. The first five amino acids of the functioning molecule of ribonuclease are:

lys – glu – thr – ala – ala

The mRNA of the gene coding for ribonuclease, for the first 15 nucleotides, has the following sequence.

AUGAAGGAAACUGCU

A genetic code, showing mRNA codons, is shown below.

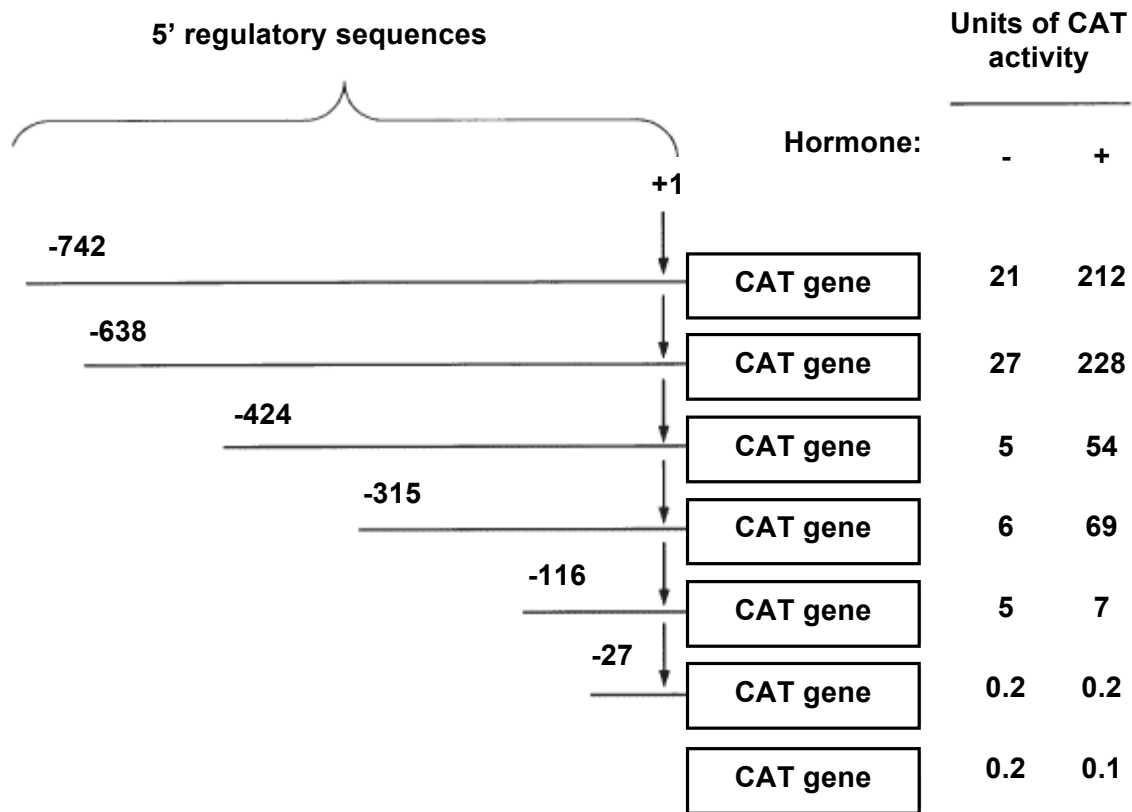
first position	second position				third position
	U	C	A	G	
U	phe	ser	tyr	cys	U
	phe	ser	tyr	cys	C
	leu	ser	STOP	STOP	A
	leu	ser	STOP	trp	G
C	leu	pro	his	arg	U
	leu	pro	his	arg	C
	leu	pro	gln	arg	A
	leu	pro	gln	arg	G
A	ile	thr	asn	ser	U
	ile	thr	asn	ser	C
	ile	thr	lys	arg	A
	met	thr	lys	arg	G
G	val	ala	asp	gly	U
	val	ala	asp	gly	C
	val	ala	glu	gly	A
	val	ala	glu	gly	G

Which event occurs to explain the information given above?

- A** The first amino acid on the polypeptide chain is removed in post-translational modification.
- B** The first codon is removed from the mRNA transcript in post-transcriptional modification.
- C** The mRNA binds to the rRNA in the second codon position.
- D** There is no tRNA with an anticodon complementary to the first codon.

- 19** Researchers studied the regulation of a hormone-responsive gene isolated 750 base pairs of DNA immediately preceding the start site of transcription (+1). They demonstrated that if these sequences are cloned upstream of the bacterial chloramphenicol acetyltransferase (CAT) gene and the DNA then introduced into mammalian cells, CAT enzyme activity increases in response to hormone treatment.

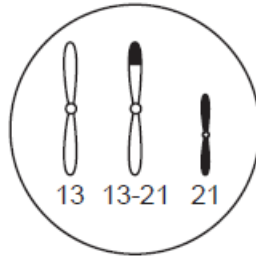
To define the sequences involved in the regulation of this gene, they made a series of deletions containing various lengths of the 5' regulatory sequences. They cloned these truncated DNA fragments upstream of the CAT gene as shown in the figure below, introduced the constructs into mammalian cells, and assayed for CAT enzyme activity in the absence (-) and presence (+) of hormone. The figure below gives the results of a representative experiment.



Assuming that there is a single responsive regulatory element in the gene, that element is located between

- A** -742 and -638
- B** -638 and -424
- C** -424 and -315
- D** -315 and -116

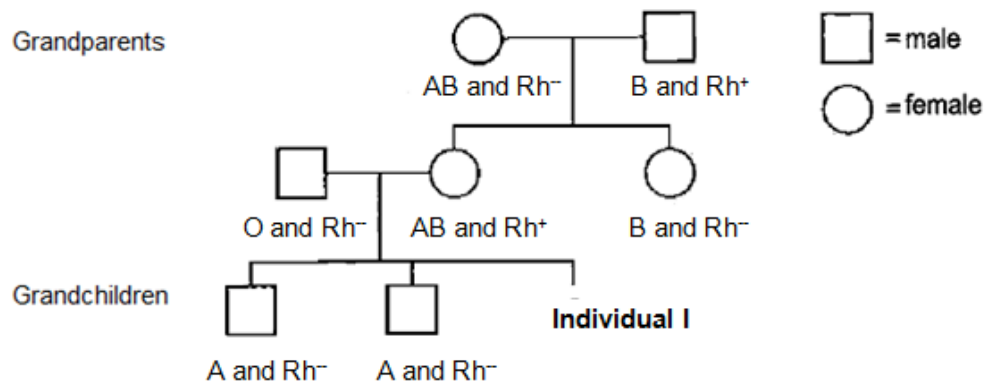
- 20** Down's syndrome can be caused by a trisomy of chromosome 21, but can also result from translocation of chromosome 21 onto chromosome 13, forming a single chromosome 13-21. The diagram shows chromosomes 13 and 21 in the nucleus of a diploid ($2n$) testis cell from a phenotypically normal male carrier of a 13-21 translocation. This cell has a chromosome number of 45.



Which is not a likely outcome of fertilisation of normal oocytes by sperm from this male?

	chromosomes in sperm	embryo
A	13 and 21	$2n = 46$; normal phenotype
B	13-21	$2n = 45$; normal phenotype
C	13-21 and 21	$2n = 46$; Down's syndrome
D	13-21 and 21	$2n = 47$; Down's syndrome

- 21** The diagram shows a family tree.



What is the probability that **Individual I** will be a girl with blood group B and Rh positive blood?

- A** 1 in 16
B 1 in 8
C 1 in 4
D 1 in 2

- 22** In a species of plant, the allele for hairy stems is dominant to the allele for smooth stems. The allele for purple stems is dominant to the allele for green stems. A plant was self-fertilised and the offspring appeared in the ratio 9 hairy, purple stem: 3 hairy, green stem: 3 smooth, purple stem: 1 smooth, green stem.

The offspring with smooth, green stems were crossed to all the other offspring.

What proportion of these crosses would be expected to produce offspring in the ratio 1 hairy, purple stem: 1 hairy, green stem: 1 smooth, purple stem: 1 smooth, green stem?

- A** 4/15
 - B** 5/15
 - C** 6/15
 - D** 9/15
- 23** Albino horses have recessive alleles which cannot be transcribed to produce pigmentation. Geneticists studying horses have found a different allele of the same gene which predisposes carriers to premature aging of the hair follicles. This allele is dominant, so, even when born with pigmented hair, the coats of horses with this allele will gradually turn white over the first 6 to 8 years of life.

Which **white** female horse would be best to use in crosses to determine the genotype of a ten-year-old, white male horse?

- A** an unrelated, ten-year-old female
- B** an unrelated, three-year-old-female
- C** a sister of the male
- D** the male's mother

- 24 A plant of genotype **SSrr** with hairy leaves and white flowers was crossed with a plant of genotype **ssRR** with smooth leaves and red flowers. The F_1 plants were allowed to self-fertilise. A χ^2 test was carried out on the results obtained for the F_2 generation.

Part of the table of values for χ^2 is shown.

degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of χ^2 in this investigation was 9.75.

Which combination correctly describes the results of the χ^2 test?

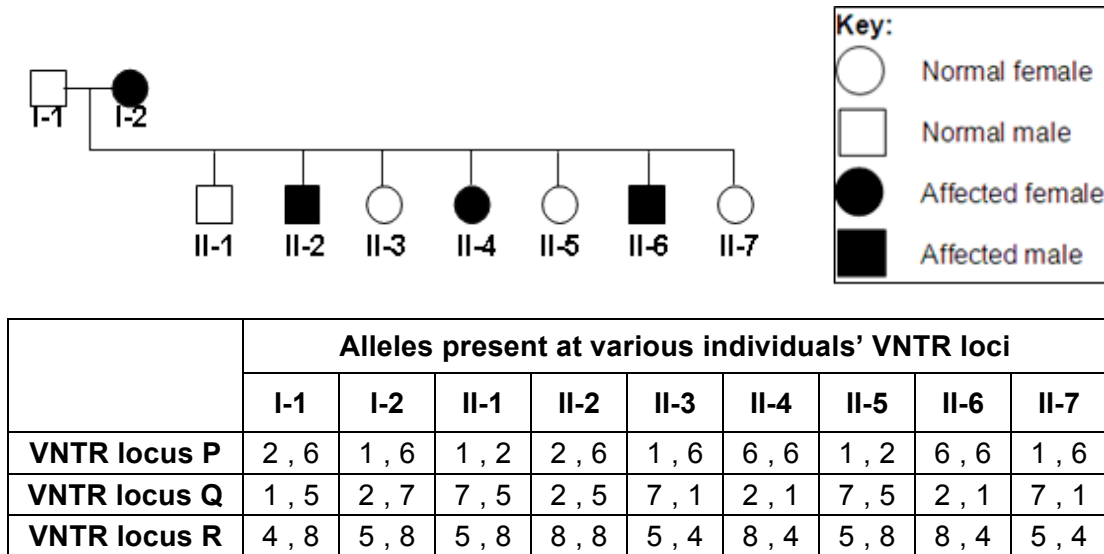
	Probability	Difference between expected and observed results	Null hypothesis
A	between 0.001 and 0.01	Significant	accept
B	between 0.001 and 0.01	Not significant	reject
C	between 0.01 to 0.05	Significant	reject
D	Between 0.01 to 0.05	Not significant	reject

- 25 Two pairs of allele govern the colour of onion bulbs. A pure-breed red strain crossed with a pure-breed white strain produces all red coloured bulbs in F_1 progeny. Self-crossing of F_1 generation produced 47 white, 38 yellow and 109 red bulbs. What does this experiment demonstrate?

- A** recessive epistasis
- B** dominant epistasis
- C** codominance
- D** there is not enough evidence to come to a conclusion

- 26** De Vivo disease is an autosomal dominant developmental disorder associated with deficiency in a type of glucose transporter. In an investigation to determine the chromosomal locus of the disease, linkage analysis of the glucose transporter gene was carried out on members of one family. Three separate VNTR loci, **P**, **Q** and **R**, located on separate chromosomes, were used.

The results of the linkage analysis are shown in the figure below. Different alleles present at various individuals' VNTR loci are denoted using different numbers.

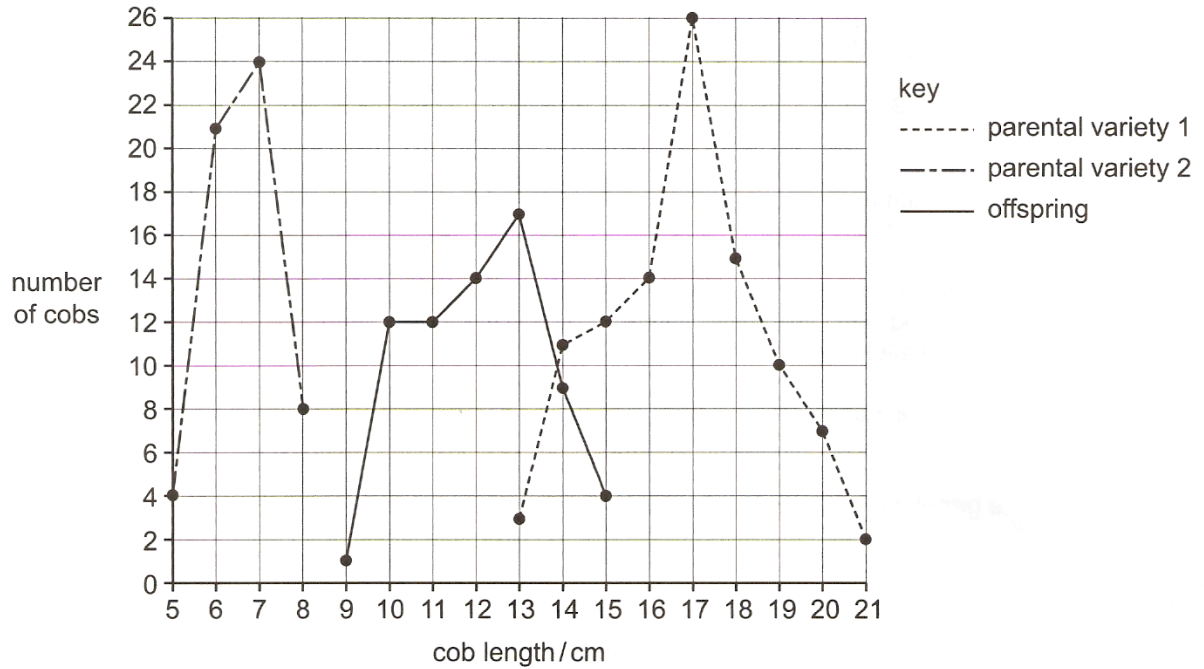


Which of the following can be concluded from the figure?

- A** Individual **I-1** is unaffected as he does not possess two copies of allele 5 for VNTR locus **Q**.
- B** Individual **II-2** is affected as he inherited two copies of allele 8 for VNTR locus **R**.
- C** Individual **II-5** is unaffected as she did not inherit allele 2 for VNTR locus **Q**.
- D** Individual **II-6** is affected as he inherited two copies of allele 6 for VNTR locus **P**.

- 27** Two pure-bred lines of two varieties of maize were crossed. The length of each of the cobs by the two parental varieties and their offspring were measured. The number of cobs in each length category was counted.

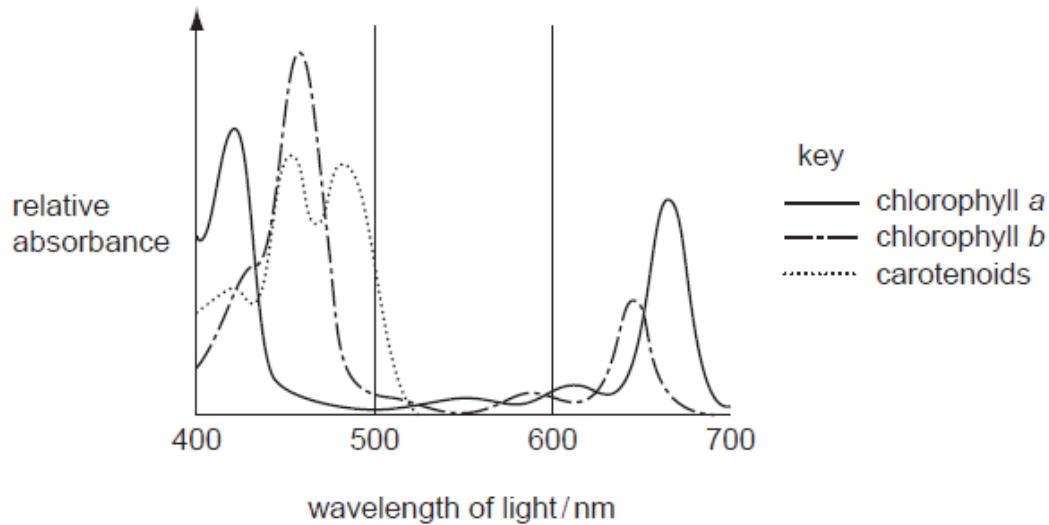
The graph shows the results.



What is the cause of the phenotypic variation in cob length within the two parental varieties and their offspring?

- A** additive effect of different genes
- B** various environmental factors
- C** linkage and crossing-over at meiosis
- D** segregation and independent assortment of alleles

28 The graph shows the absorption spectra of some pigments found in chloroplasts.

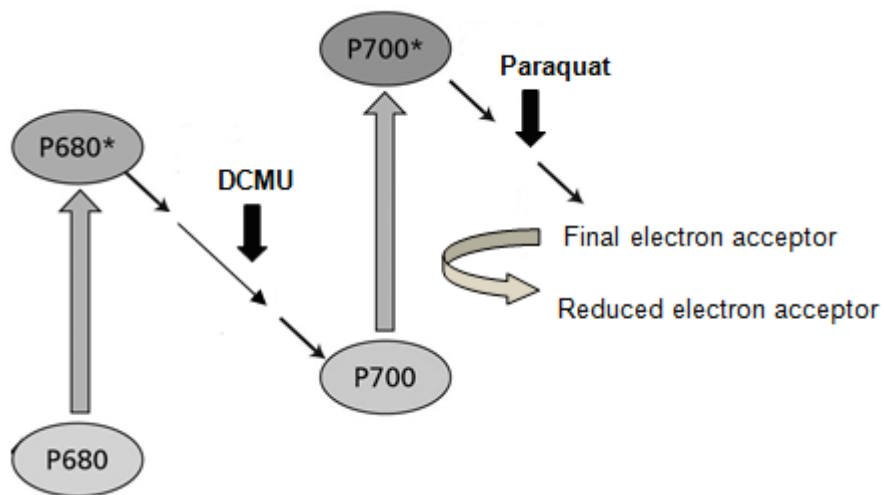


Which statements are correct?

1. Having several pigments, rather than one, increases the efficiency of photosynthesis.
2. Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
3. Photosynthesis will be fastest in light at the red end of the spectrum, as red light has higher energy than blue light.
4. Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.

- A** 1 and 2 only
- B** 1 and 3 only
- C** 1, 2 and 4 only
- D** 2 and 4 only

- 29 The diagram shows the electron transport chain embedded in thylakoid membranes of chloroplasts.

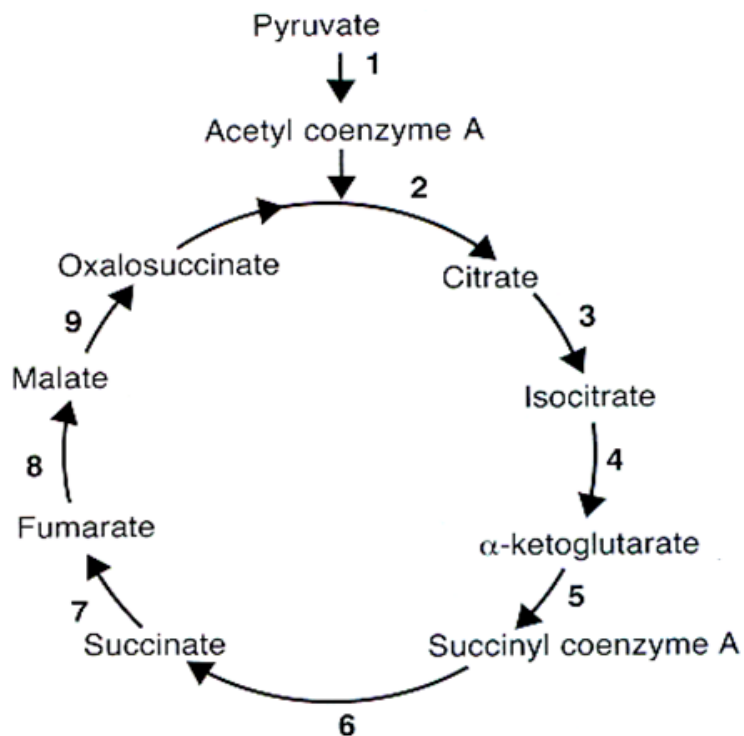


DCMU and paraquat are inhibitors of electron transport chain at the positions indicated in the diagram. These two inhibitors are added separately to isolated chloroplasts.

Which of the following correctly represents the results in the presence of DCMU and paraquat?

	DCMU only			Paraquat only		
	NADPH	Oxygen	ATP	NADPH	oxygen	ATP
A	Present	Absent	Present	Absent	Absent	Present
B	Absent	Present	Absent	Absent	Present	Present
C	Absent	Present	Present	Present	Present	Absent
D	Absent	Absent	Absent	Absent	Absent	Present

- 30 The diagram below shows the Krebs Cycle.



At which stages are NAD reduced?

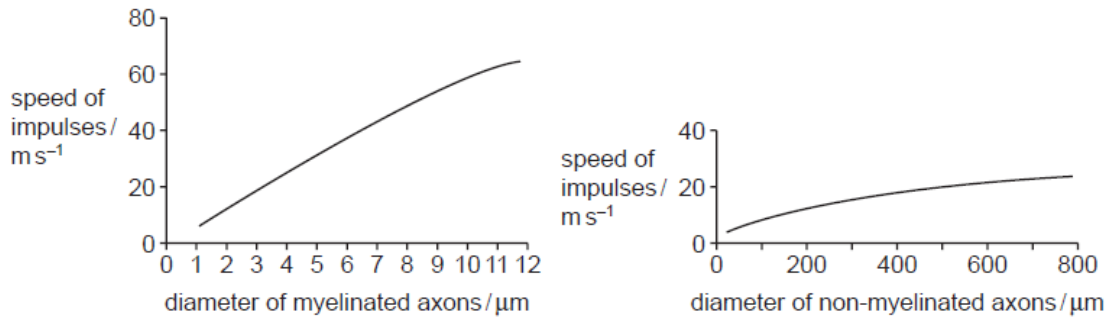
- A** 1, 4, 5 and 7
B 1, 4, 5 and 9
C 2, 4, 5 and 7
D 4, 5, 7 and 9
- 31 A person has malfunctioning β -cells in the islets of Langerhans. What will be the levels of glucose, insulin and glucagon in the blood after that person has eaten a meal rich in carbohydrates?

	glucose	insulin	glucagon
A	low	high	low
B	high	low	low
C	low	high	high
D	high	low	high

- 32** The ability of organisms to respond rapidly to stimuli is limited by the speed of the impulses in their neurones.

The axons of invertebrate neurones lack a myelin sheath. The axons of most vertebrate neurones are myelinated.

The graphs show the speed of impulses in these two types of axon.

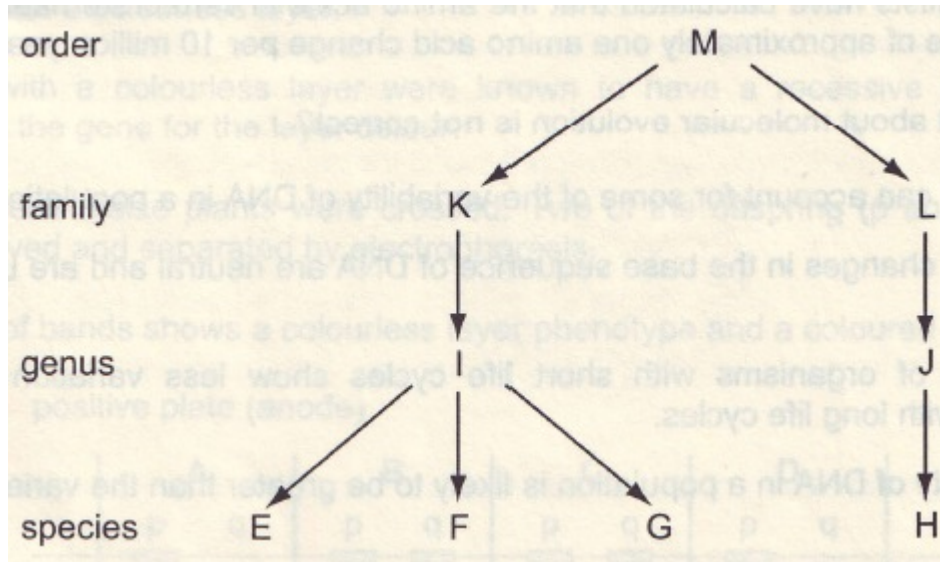


Which statements about these data are correct?

1. The action potential in myelinated axons is greater than the action potential in non-myelinated axons.
2. The speed of impulses is changed by the diameter of the axon.
3. Increasing the diameter of a myelinated axon causes a greater change to the conduction speed than increasing the diameter of a non-myelinated axon.
4. The presence of myelin increases the speed at which impulses are conducted.

- A** 1, 2 and 3 only
B 1, 2 and 4 only
C 2, 3 and 4 only
D 3 and 4 only

- 33 The diagram shows part of a phylogenetic classification of four species into taxonomic groups.



Which row correctly describes the order M in comparison with genera I and J?

	relative size of taxonomic groups	number of similarities between members of the taxonomic groups	time since members of taxonomic groups shared a common ancestor
A	M is larger than I or J	fewer similarities between members of M than between members of I or J	members of M shared a common ancestor a longer time ago than those of I or J
B	M is larger than I or J	more similarities between members of M than between members of I or J	members of M shared a common ancestor a shorter time ago than those of I or J
C	M is smaller than I or J	fewer similarities between members of M than between members of I or J	members of M shared a common ancestor a shorter time ago than those of I or J
D	M is smaller than I or J	more similarities between members of M than between members of I or J	members of M shared a common ancestor a longer time ago than those of I or J

- 34** The herbaceous plant *Achillea lanulosa* is widespread in the Northern Hemisphere. Phenotypic variation in the species has been extensively studied along an altitudinal gradient from sea level to over 3,000 metres. In California, there is conspicuous variation in height; alpine plants are only several centimetres tall while those in the San Joaquin Valley may reach a height of 1.8 metres. Adaptation to different local environments also results in variation in physiological processes such as photosynthetic rate, resistance to cold, and timing of dormancy.

Which of the following methods would be best to determine whether the phenotypic variation in *Achillea lanulosa* is due to genotypic variation?

- A** Assessing the amount of phenotypic variation within populations along the entire altitudinal range
- B** Determining whether phenotypic differences are maintained when plants from different altitudes are grown under the same environmental conditions
- C** Determining whether viable hybrids between phenotypically different populations can be produced
- D** Determining whether hybrids between phenotypically different populations grow at altitudes intermediate between the parent populations

- 35** The following statements relate to molecular phylogenetics.

1. Lines of descent from a common ancestor to present-day organisms have undergone similar, fixed rates of DNA mutation.
2. Organisms with similar base sequences in their DNA are closely related to each other.
3. The number of differences in the base sequence of DNA of different organisms can be used to construct evolutionary trees.
4. The proportional rate of fixation of mutations in one gene relative to the rate of fixation of mutations in other genes stays the same in any given line of descent.

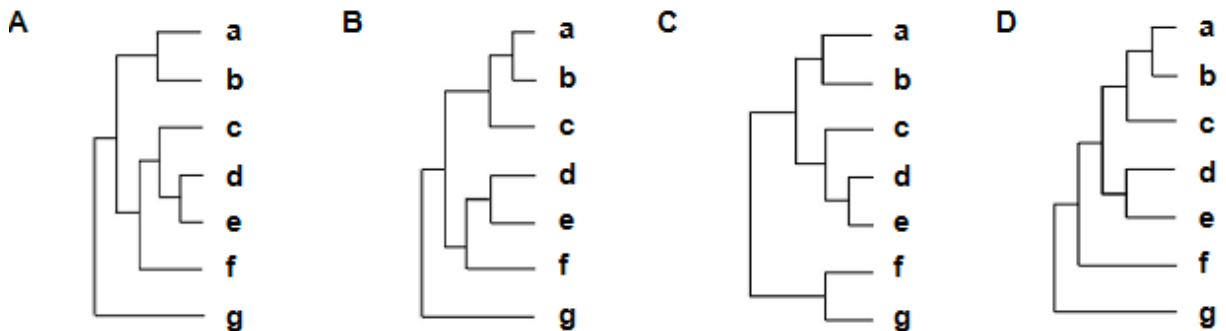
Which statements, when taken together, suggest the existence of a 'molecular clock' that enables scientists to estimate the time at which one species might have diverged from another?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 36 The table shows the number of estimated nucleotide substitutions that have occurred since the divergence of seven species **a** to **g**.

	b	c	d	e	f	g
a	39	72	128	126	159	269
b		81	130	128	158	268
c			129	127	157	267
d				56	154	271
e					151	268
f						273

Which of the following phylogenetic trees best shows the relationship among these seven species?



- 37 In obtaining the gene that codes for human actin, reverse transcriptase was used.

Why is reverse transcriptase **not** used for obtaining the gene that produces a human sodium ion channel protein?

- A** Reverse transcriptase is a viral enzyme, and in the current environment of suspicion of viruses, it is not possible to use such enzymes.
- B** The amino acid sequence for this protein is not known, so this is a gene that has not yet been located by the human genome project.
- C** The antibiotic resistance genes transferred by reverse transcriptase with the insulin gene may be transferred to pathogens.
- D** The sodium ion channel protein gene is expressed at low levels in human cells, so the mRNA produced is swamped by other mRNA.

- 38** A gene sequence to be cloned is shown below. N_x denotes an unspecified sequence.

5' – AATCGCCGAGCTCGGTTACG(N_x)TAGGCATCTCGAGCCTAGG – 3'

Which pair of primers will be effective for PCR amplification of the gene?

P: 5' AATCGCC 3'

Q: 5' ATCCGTA 3'

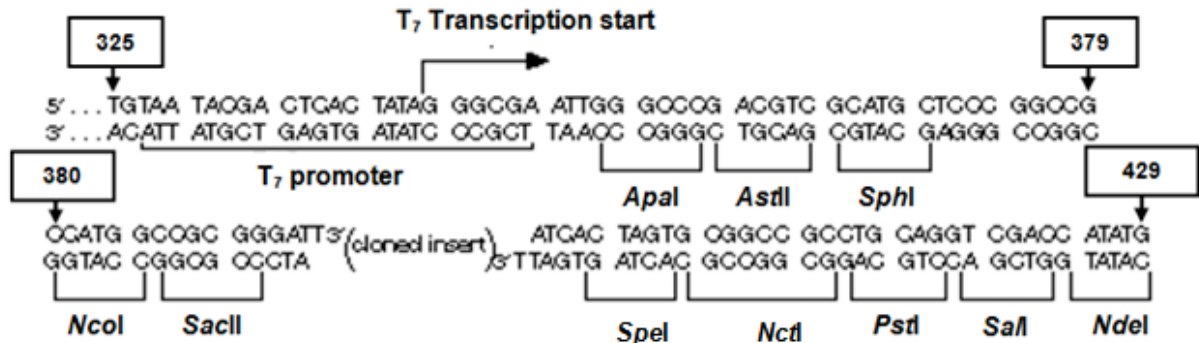
R: 5' TAGGCAT 3'

S: 5' TTAGCGG 3'

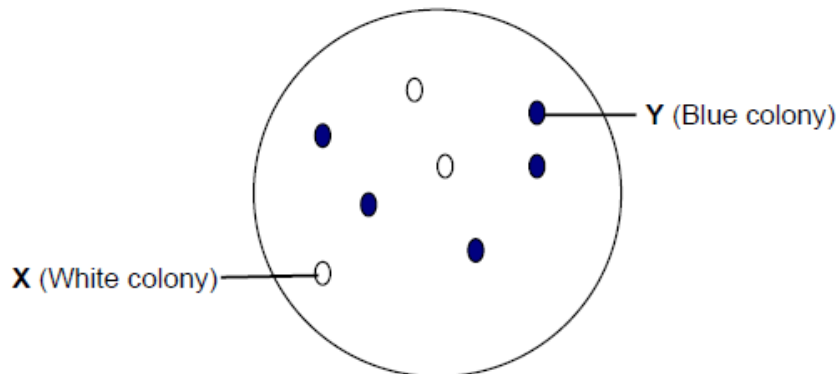
T: 5' CCTAGGC 3'

- A** P and Q
- B** P and R
- C** P and T
- D** Q and S

- 39 pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, *Tet^R* and *lacZ*. The figure below shows a portion of the multiple cloning site (MCS) of the cloning vector. It is located within the *lacZ* gene.



Using *SacII* as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS shown above, and the recombinant plasmid is transformed into *E. coli* cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



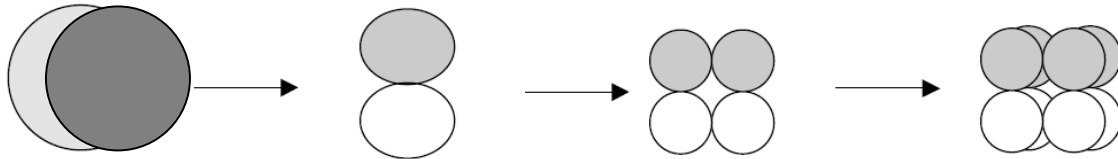
Colonies X and Y are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *SacII*. Which of the following correctly shows the fragments obtained?

	Colony X	Colony Y
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.25kb and 0.34kb fragments

- 40** When a fertilized egg begins development, the outcome of the first three cell cycles is shown as follows:



The localization of a specific protein during each of the above four early developmental stages is examined. This protein is required for the formation of ectoderm. The location of the protein is indicated by grey shading.



At the four-cell stage, are the four cells equivalent in terms of the different fates their daughter cells can take on, and why?

	Yes or No to the four-cell equivalence in terms of their fates	Justification
A	No	The protein required for ectoderm formation is found only in the upper two cells; therefore, only these two cells can become ectoderm. The lower two cells can continue to differentiate to become mesoderm or endoderm at this point, but cannot become ectoderm.
B	No	The protein required for ectoderm formation is found only in the upper two cells. These proteins can be transported to the other two lower cells such that the mesoderm or endoderm can be formed.
C	Yes	The protein required for ectoderm is needed for differentiation of the other germ layers since the ectodermal cells still remain totipotent and can take on any developmental pathway.
D	Yes	The protein required for ectoderm is needed only by the upper two cells since the lower two cells can differentiate to become mesoderm or endoderm at this point.

2013 H2 Biology Preliminary Examination Paper 1 ANSWER KEY

1	B	2	A	3	C	4	C	5	D
6	B	7	B	8	D	9	A	10	D
11	A	12	B	13	D	14	D	15	A
16	B	17	C	18	A	19	D	20	D
21	B	22	A	23	A	24	C	25	A
26	C	27	B	28	C	29	D	30	B
31	B	32	C	33	A	34	A	35	B
36	D	37	D	38	C	39	A	40	A